Chemistry of the Earth: Lecture 9, Water

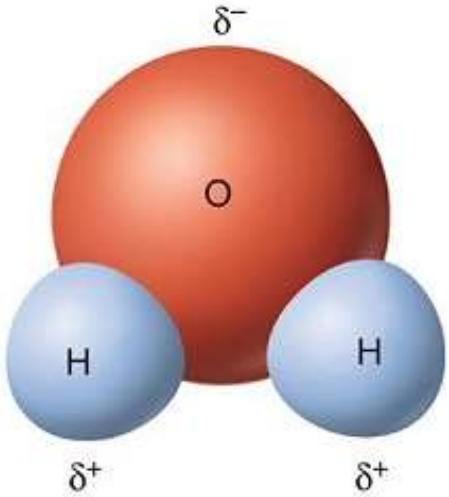


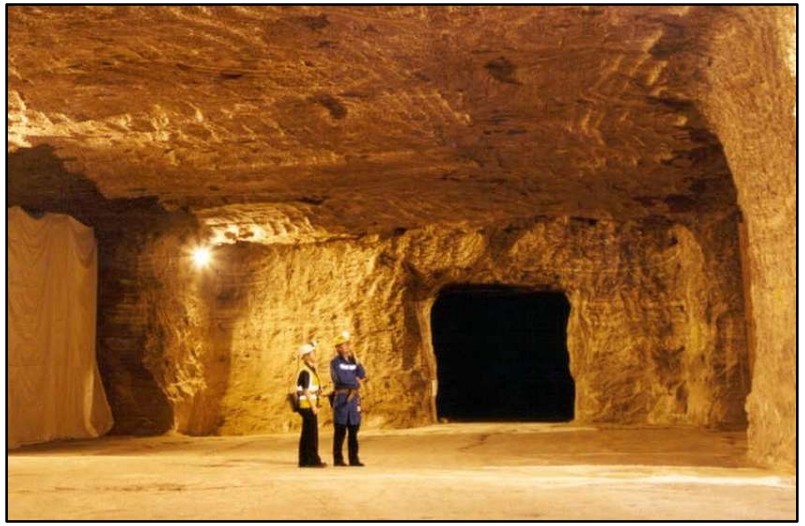
Part 1: Global Meteoric Water

OBJECTIVES OF THIS LECTURE AND PRACTICAL

- Investigation of how stable isotopes are used to understand global meteoric water and palaeoclimate
- Examine how to describe the chemistry of water
- Investigate geochemical weathering reactions involving rocks and water







READING and RESOURCES

	Water and climate	Chemistry of Water
Grotz & Jordan (Recommended)	**Ch. 12, p. 350-354**	**Ch 16, p. 454 – 461**
Misra: (Recommended)	**Ch. 11.4**	**Ch. 7 (limited sections)**
Gill: (Recommended)	**Ch. 4, p. 60-69**	**Ch. 10, p. 197-200**
White:	Ch. 9.6	Ch. 6
McSween:	Ch. 13, p. 271-273	Ch.4, Ch.7

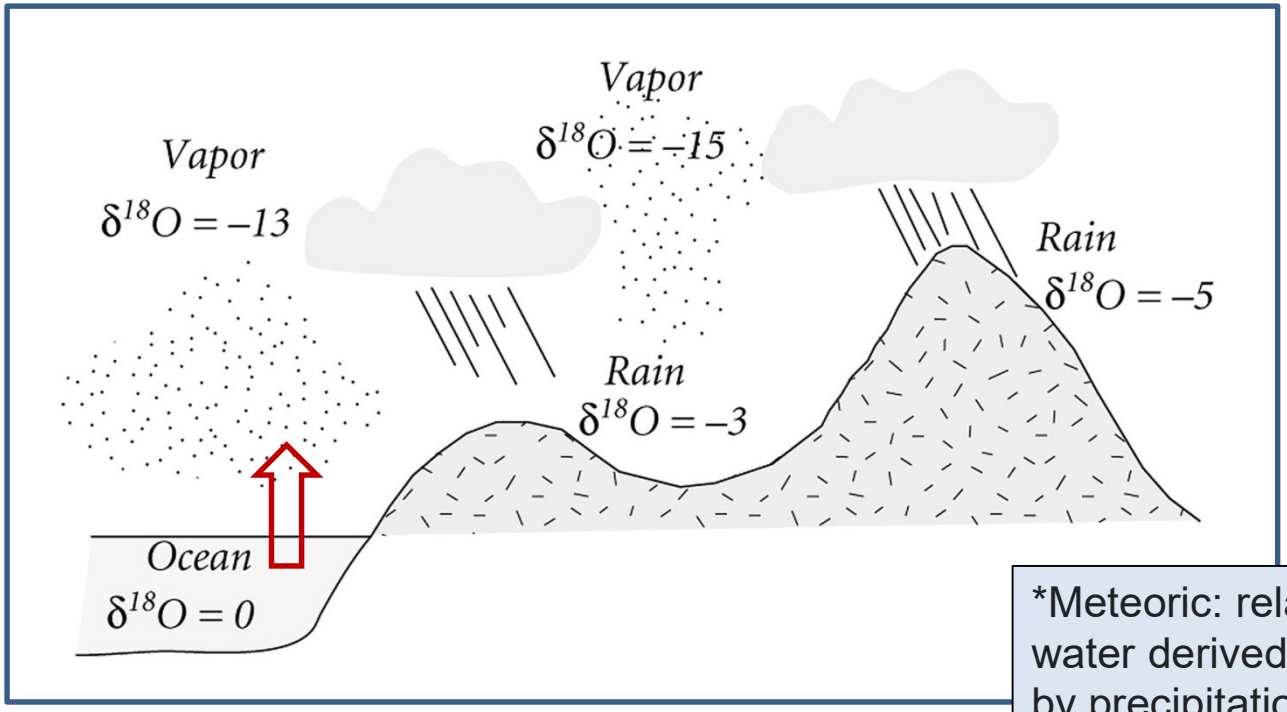
Global Meteoric* Water

Evaporation and Condensation of Water: Oceans and Rain

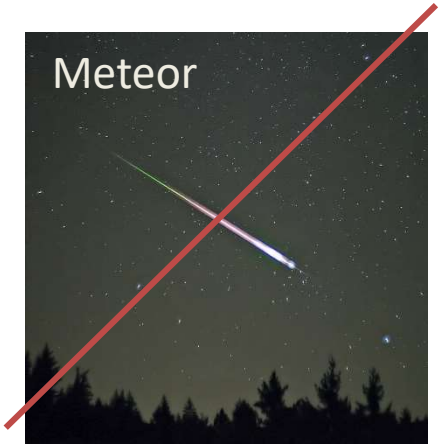
Water, H_2O. Isotopes: ^{1}H, ^{2}H (=D), ^{16}O, ^{17}O, ^{18}O. Nine isotopologues.

Ocean water: The ocean is well mixed and has a uniform isotopic composition. Mean values are very close to SMOW, i.e. δD and $\delta^{18}O$ are 0 ‰.

The ocean is an open system with respect to water vapour, which escapes to the atmosphere as soon as it forms.



White Fig. 9.10



Wikipedia / Navicore

*Meteoric: relating to or denoting water derived from the atmosphere by precipitation or condensation

Evaporation of ocean water:

Equilibrium isotopic fractionation factors at 25 °C (298.15 K) are:

For oxygen, $\alpha_{ow\text{-}wv} = \frac{(^{18}O/^{16}O)_{ow}}{(^{18}O/^{16}O)_{wv}} = 1.0092$

For hydrogen, $\alpha_{ow\text{-}wv} = \frac{(D/H)_{ow}}{(D/H)_{wv}} = 1.074$

ow = ocean water
wv = water vapour

Leading to $\delta^{18}O_{wv} = -9.12$ ‰

$\delta D_{wv} = -68.90$ ‰

- **Ocean water is enriched in heavy isotopes relative to water vapour.**
- In addition, kinetic isotope effects contribute to making the water vapour even lighter.

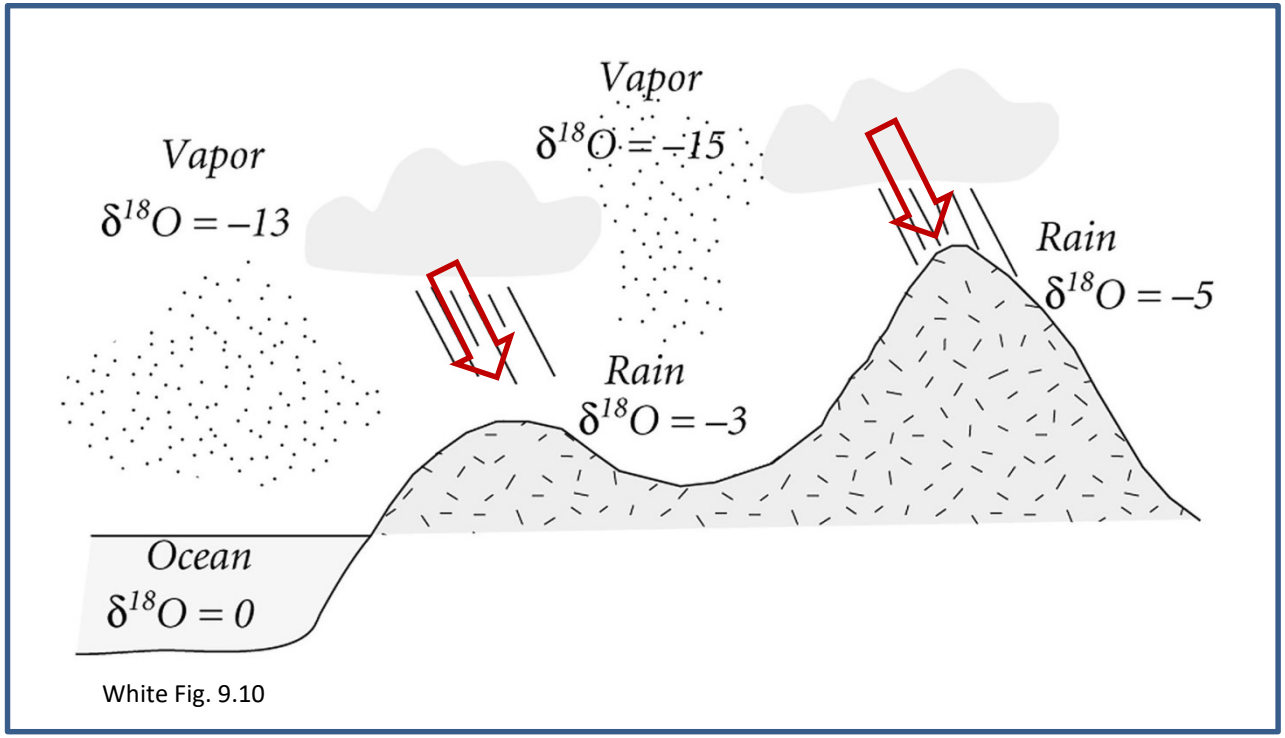
Condensation of water vapour:

Condensation of water vapour in a cloud is an equilibrium process.

Condensation is the reverse of evaporation.

Water vapour in the atmosphere is not a huge reservoir like the ocean, so its isotopic composition changes as water condenses.

The first raindrops have higher $\delta^{18}O$ than the vapour: this makes the vapour lighter, i.e. enriched in $\delta^{16}O$ and δD.



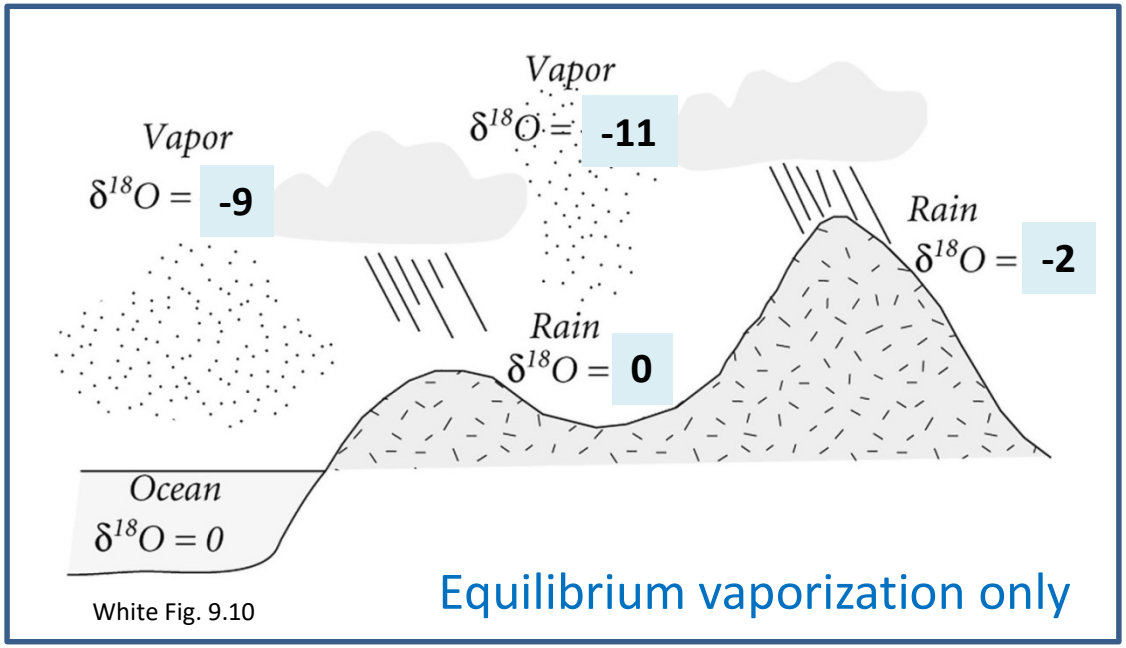
White Fig. 9.10

The next batch of raindrops is lighter than the first rain, and so on. The vapour also becomes lighter.

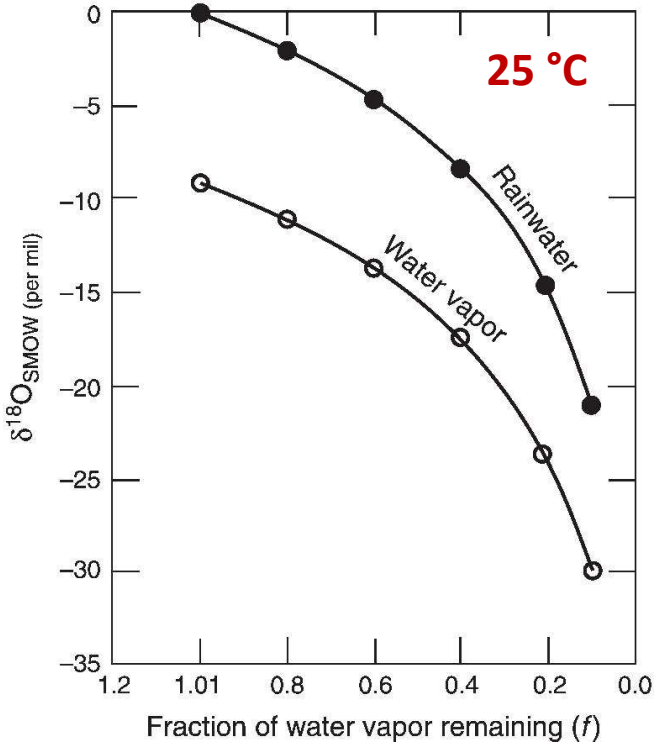
Calculation of $\delta^{18}O$ in water vapour and rainwater, assuming only equilibrium vaporization of the ocean (See Misra Example 11-3 p. 264)

Fraction of water vapor remaining (f_{wv})	$\delta^{18}O_{wv}^{remaining}$ (per mil)	$\delta^{18}O_{rw}$ (per mil)	Fraction of water vapor remaining (f_{wv})	$\delta^{18}O_{wv}^{remaining}$ (per mil)	$\delta^{18}O_{rw}$ (per mil)
1.0	–9.1	0.0	0.5	–15.4	–6.4
0.9	–10.1	–1.0	0.4	–17.4	–8.4
0.8	–11.2	–2.1	0.3	–20.0	–11.0
0.7	–12.4	–3.3	0.2	–23.7	–14.7
0.6	–13.8	–4.7	0.1	–29.9	–21.0

Table 11.4 Progressive change in $\delta^{18}O$ (water vapor) and in the corresponding $\delta^{18}O$ (rainwater) at 25°C (298.15 K) with increasing precipitation.



White Fig. 9.10



Misra Fig. 11.4

Meteoric water – water derived from precipitation (rain, snow, etc)

In addition to distance from the ocean, other effects change the isotopic composition of rain:

Isotopic fractionation gets larger with decreasing temperature.

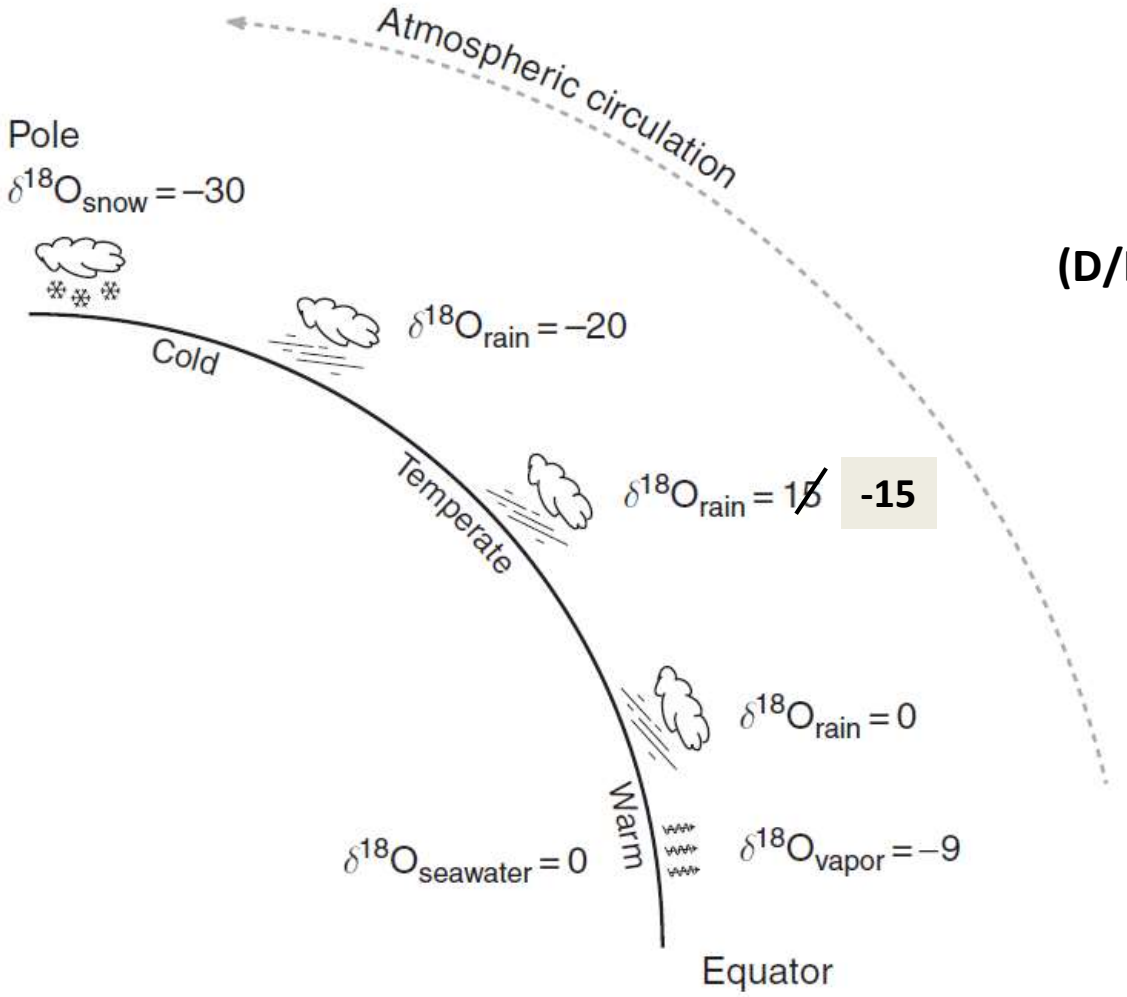
Hence, $\delta^{18}O$ and δD values of meteoric water vary with:

- latitude - hot at equator, cold at poles
- elevation - warm at sea level, cold at tops of mountains

At high latitude and high elevation, meteoric water is more depleted in ^{18}O and D.

$\delta^{18}O$ = -50 ‰ and δD = -450 ‰ have been measured at the South Pole.

Variation of $\delta^{18}O$ with latitude



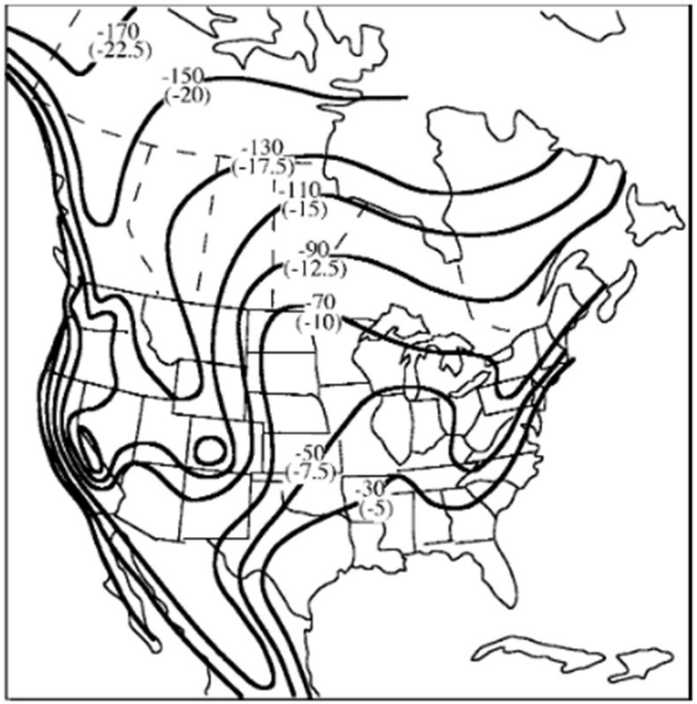
(D/H also varies with latitude)

Albarede Fig. 7-9

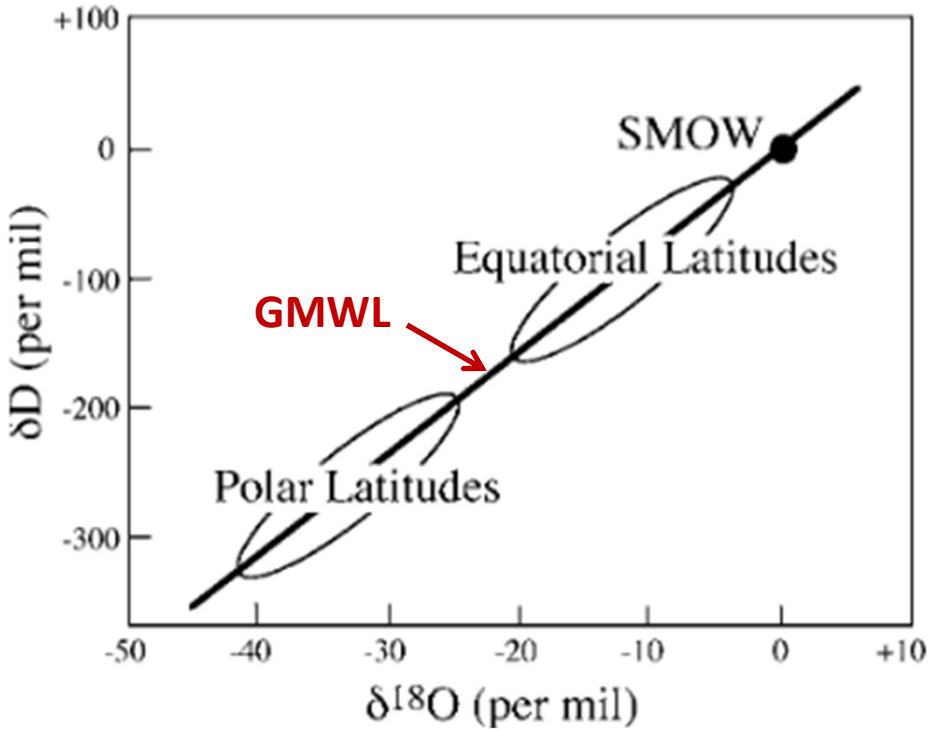
Global meteoric water line (GMWL)

Global Meteoric Water Line: $\delta D = 8\,\delta^{18}O + 10$

Meteoric waters around the Earth have isotopic compositions that lie on or close to this line.



Contours of δD and $\delta^{18}O$ (in parentheses) in rainwater in the USA
McSween Fig. 3-6



δD vs $\delta^{18}O$ in rainwater around the world
McSween Fig. 3-7

Nb: On p. 272 of McSween, the equation for the Meteoric Water Line is incorrect.

Relating Global Meteoric Water to Palaeoclimatology

During a glacial period, ice sheets lock up a disproportionate amount of ^{16}O, leaving the oceans relatively enriched in $\delta^{18}O$.

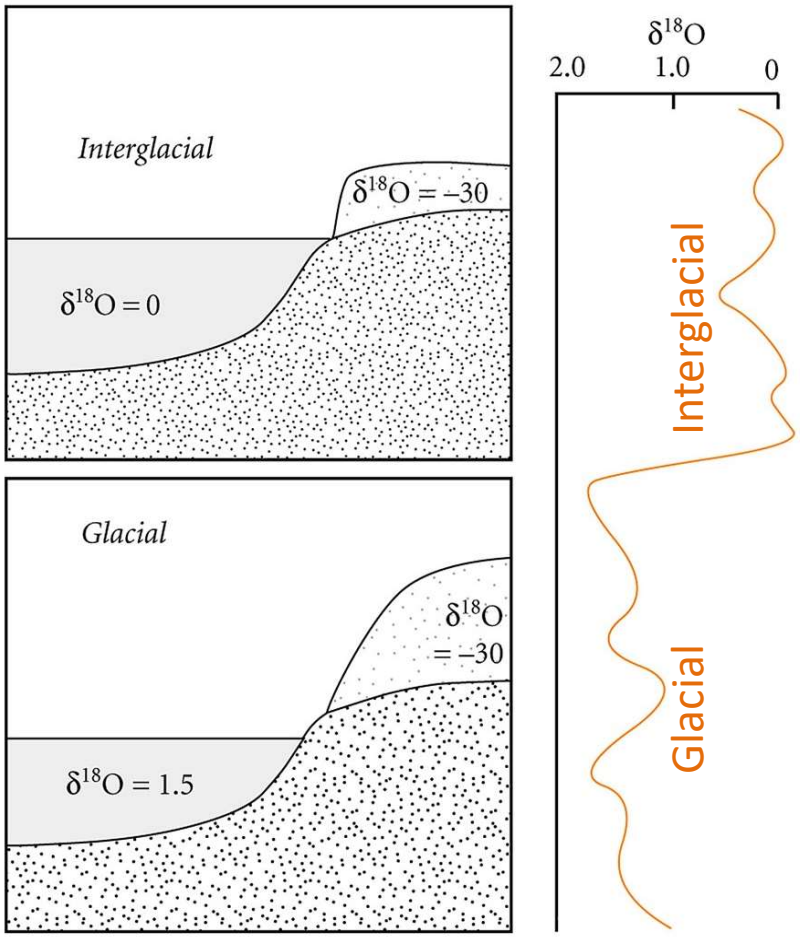
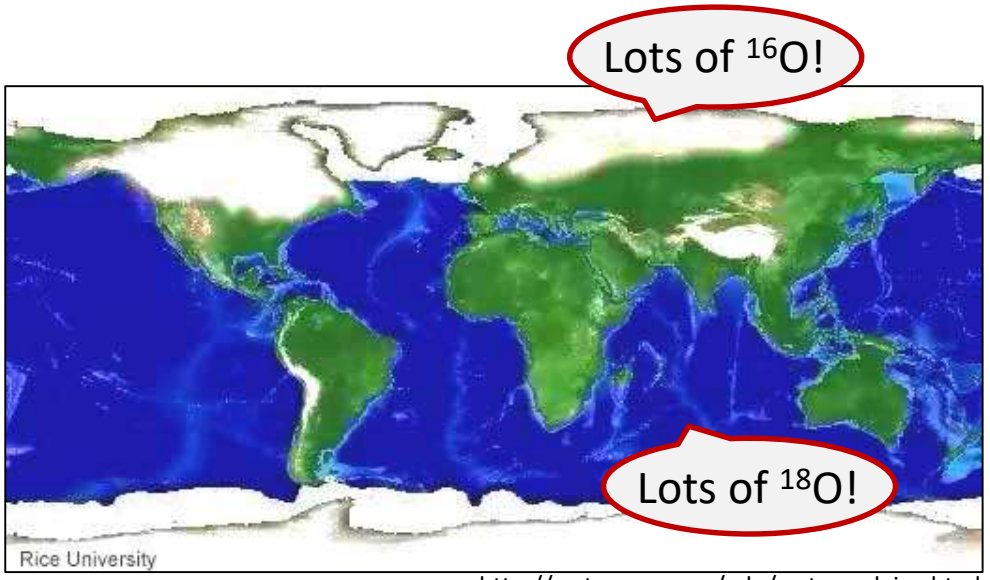


Figure 9.22 Cartoon illustrating how $\delta^{18}O$ of the ocean changes between glacial and interglacial periods.



http://water.usgs.gov/edu/watercycleice.html

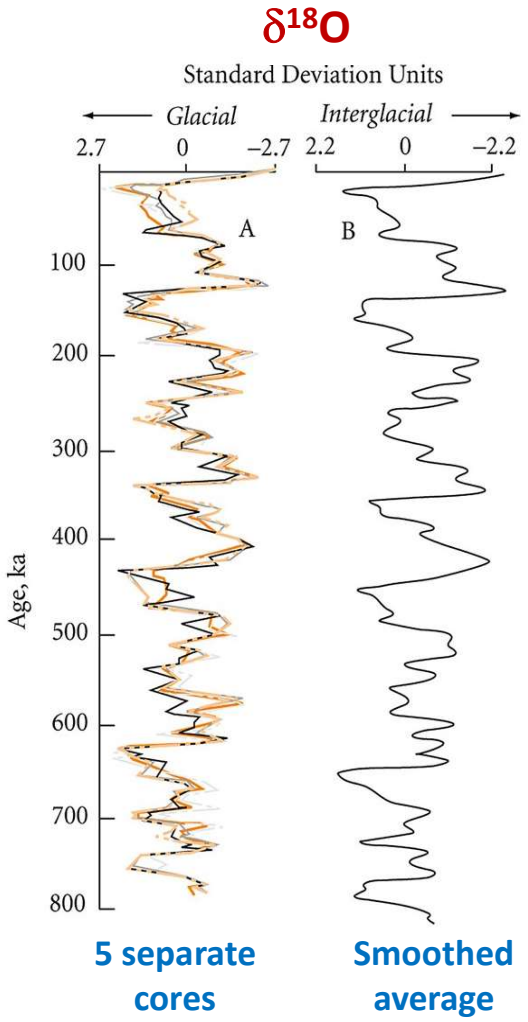
Extent of major glaciers at the height of the last ice age (18,000 years ago)

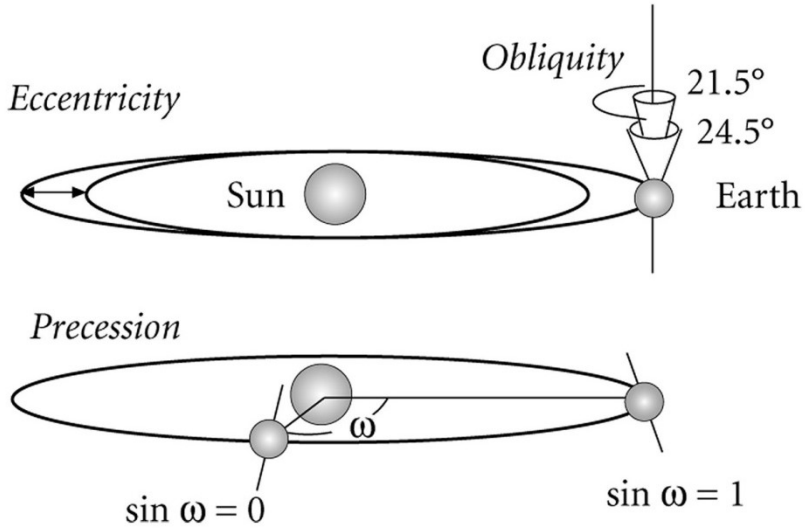
White, Fig. 9.22

Milankovitch Cycles

Deep-sea sediment cores show changes in oxygen isotope compositions that show regular periodicities.

Periodicity is related to changes in the Earth's orbit around the Sun (Milankovitch cycles) and resulting variations in the solar energy flux (insolation).

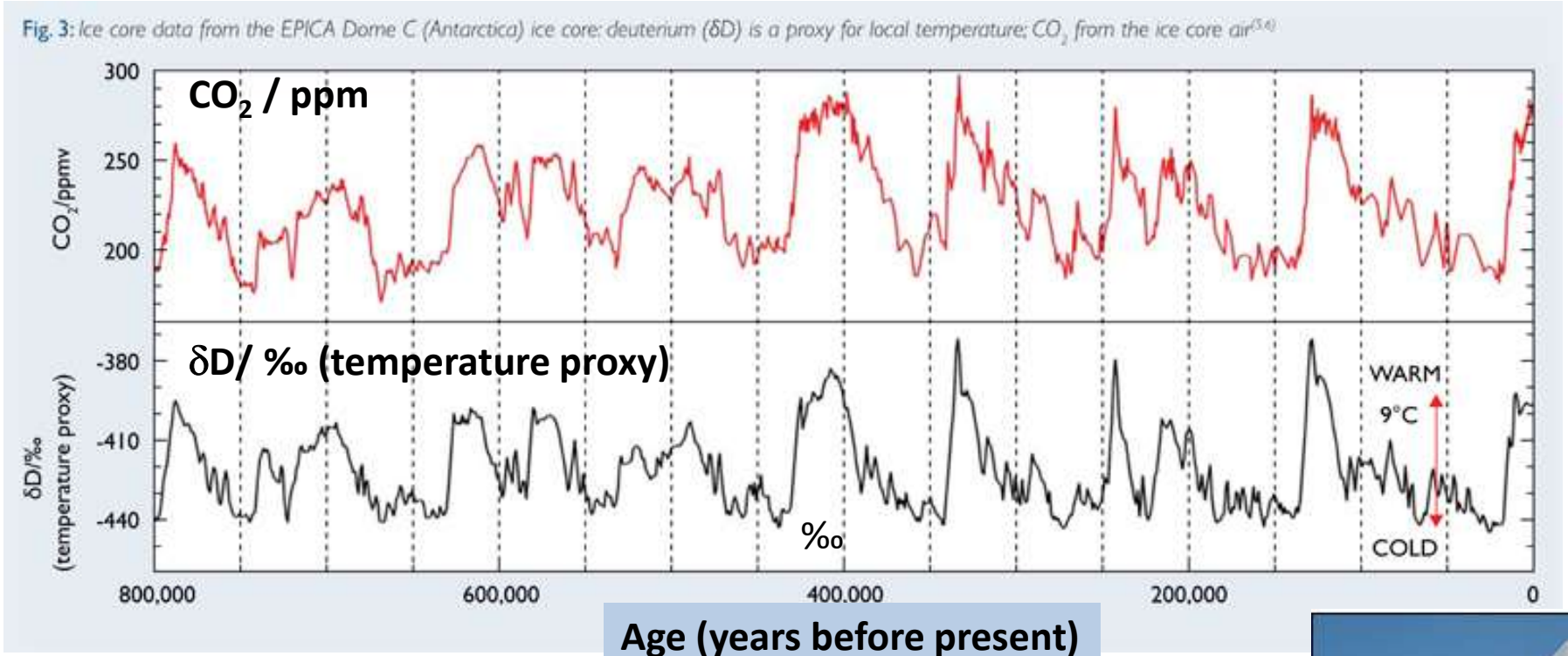




Orbital parameter	Periodicity
Eccentricity	123,000 years
	85,000 years
	58,000 years
Tilt	41,000 years
Precession	23,000 years
	18,000 years

Figures from White

Ice cores in Greenland and Antarctica preserve an important record of climate change



https://www.bas.ac.uk/data/our-data/publication/ice-cores-and-climate-change/



The oldest continuous ice core records to date extend 123,000 years in Greenland and 800,000 years in Antarctica.

http://earthobservatory.nasa.gov/Features/Paleoclimatology_IceCores/

Chemistry of the Earth: Lecture 9, Water

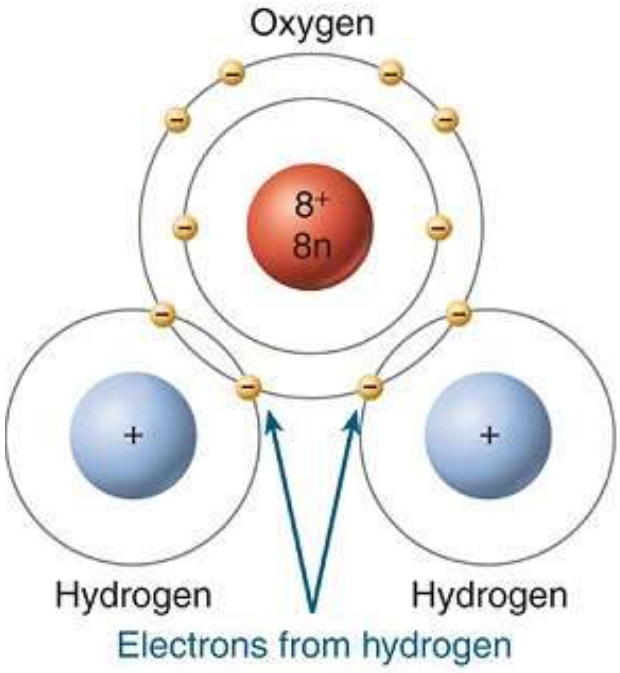


Part 2: Water Chemistry

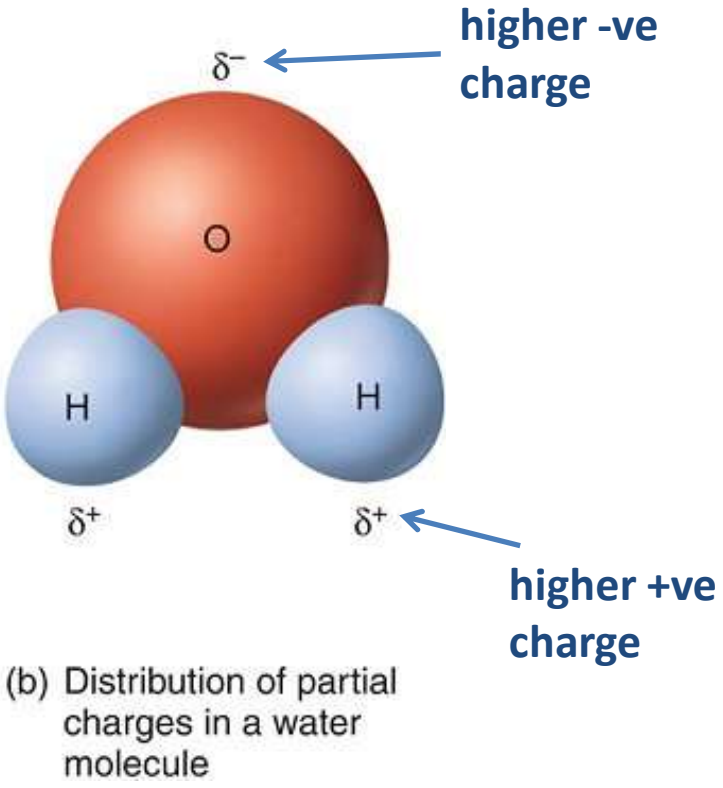
CHEMISTRY OF WATER

Liquid water is a very important component of geochemical systems – atmosphere, hydrosphere, biosphere, weathering, formation and alteration of sedimentary, igneous, and metamorphic rocks.

Why? - Liquid water is a **POLAR** solvent.
This means the charge is not distributed evenly over the water molecule.



(a) Electron shells in a water molecule



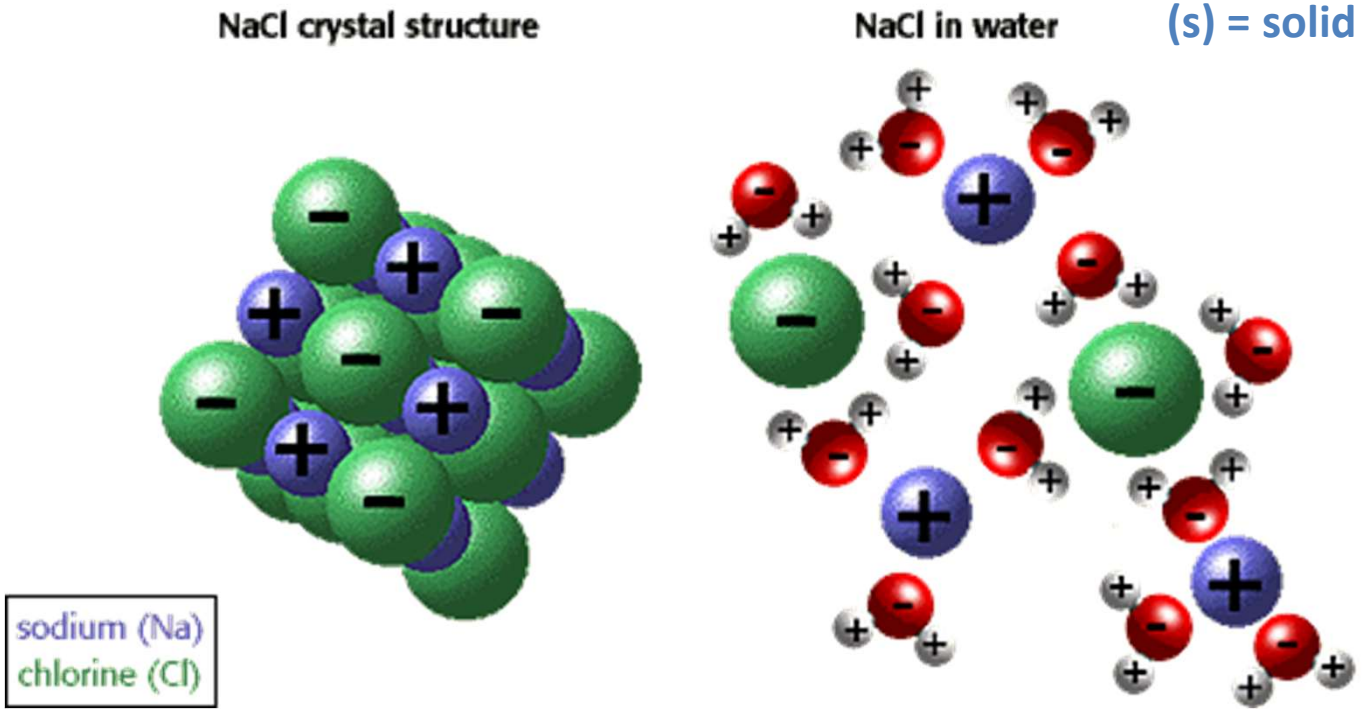
(b) Distribution of partial charges in a water molecule

Misra Ch7.1

When a **salt** dissolves in water, ions are surrounded by water molecules that orientate themselves with the opposite charge. This produces a layer of water that keeps the dissolved ions apart.

Example, halite: $NaCl_{(s)} = Na^{+}_{(aq)} + Cl^{-}_{(aq)}$

(aq) means the ion is in solution
(s) = solid



Water is the **SOLVENT**, NaCl is the **SOLUTE**.
A **SOLUTION** is the combination of both.

http://butane.chem.uiuc.edu/pshapley/genchem1/l21/1.html

Salts, Acids and Bases

Because the dissolved ions are charged particles, water can conduct an electric current. Aqueous solutions are therefore called **ELECTROLYTE** solutions.

A **SALT** is an ionic compound that dissolves in water.
- *except* for when the cation is H^+ (e.g. HCl) or OH^- (e.g. NaOH) or O^{2-} (e.g. Na_2O)

These exceptions are made because:

H^+ compounds are **ACIDS**

OH^- , O^{2-} compounds are **BASES / ALKALIS**

GCSE Review of acids and bases:
http://www.bbc.co.uk/schools/gcsebitesize/science/add_aqa_pre_2011/ions/acidsbasesrev1.shtml

Acids

When **acids** dissolve in water they produce hydrogen ions, H^+.

For example, hydrochloric acid: $HCl_{(aq)} \rightarrow H^+_{(aq)} + Cl^-_{(aq)}$

Bases v alkalis

Bases are substances that react with acids and neutralise them. They are usually metal oxides, metal hydroxides, metal carbonates or metal hydrogen carbonates. Many bases are insoluble - they do not dissolve in water.
If a base does dissolve in water, we call it an **alkali**.

When **alkalis** dissolve in water they produce hydroxide ions, OH^-.

For example, sodium hydroxide: $NaOH_{(aq)} \rightarrow Na^+_{(aq)} + OH^-_{(aq)}$

Neutralisation reaction: Acid plus Alkali

When the H^+ ions from an acid react with the OH^- ions from an alkali, a neutralisation reaction occurs to form water.

This is the equation for the reaction: $H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(l)}$

(l) = liquid

If you look at the equations above for sodium hydroxide and hydrochloric acid, you will see that there are Na^+ ions and Cl^- ions left over. Put together, these form sodium chloride, NaCl.

$$HCl_{(aq)} + NaOH_{(aq)} \rightarrow H^+_{(aq)} + Cl^-_{(aq)} + Na^+_{(aq)} + OH^-_{(aq)} \rightarrow NaCl + H_2O$$

A **salt** is an ionic compound that results from the neutralization reaction of an acid and a base.

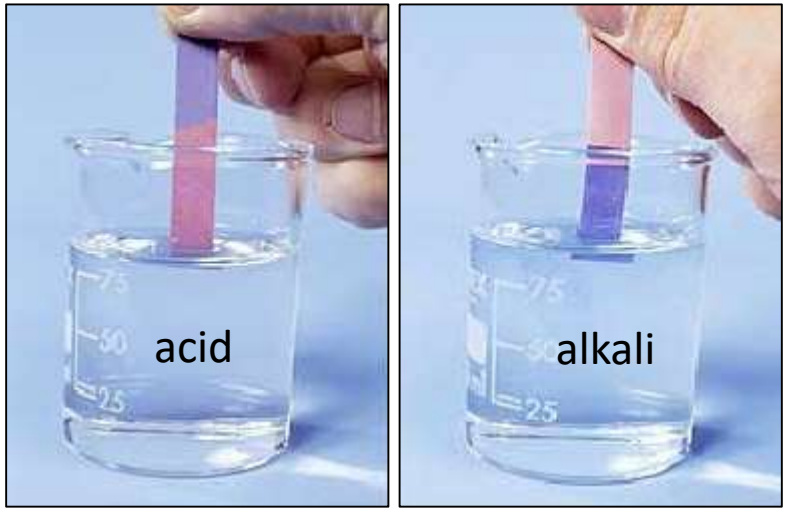
Acid + Alkali = Salt + Water

Acid + Carbonate = Salt + Water + Carbon dioxide

Acid + Metal = Salt + Hydrogen

Acidity and the pH scale

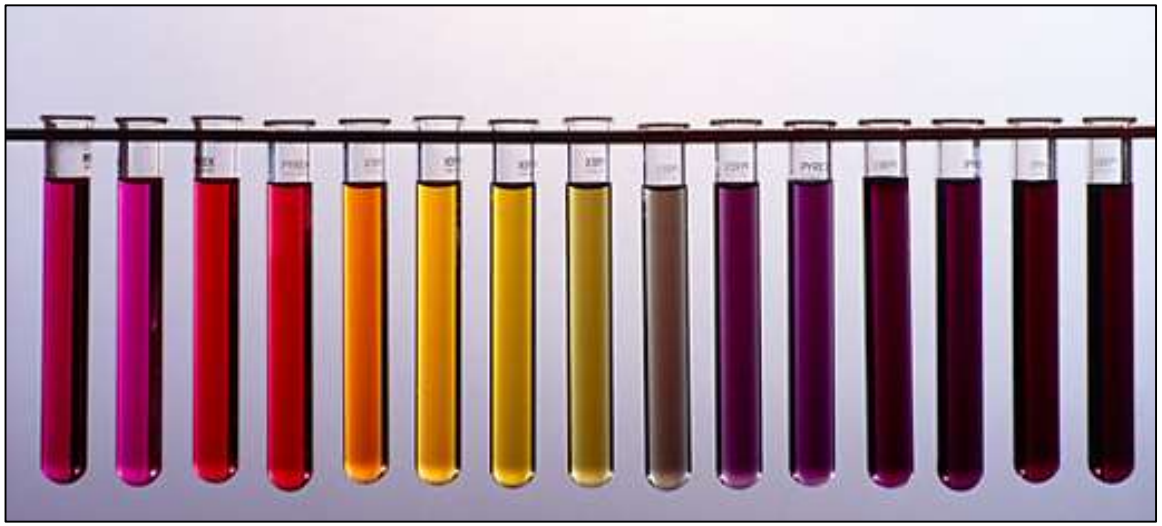
Litmus test



acidic solution alkaline solution

- When an acid is dissolved in water, we get an **acidic** solution
- When an alkali is dissolved in water, we get an **alkaline** solution
- If a solution is neither acidic nor alkaline, we call it **neutral**

(Bitesize – Key Stage 3)



pH 0 pH 7 pH 14

← more acidic more alkaline →

Universal Indicator:
a measure of pH

pH 0 – strongly acidic
pH 7 – neutral
pH 14 – strongly alkaline

The pH scale

Acidity is a measure of the concentration of H^+ ions (protons) in solution.

pH means "potential" Hydrogen.

$pH = -\log_{10} [H^+]$, where $[H^+]$ is the concentration of H^+ ions in mol kg^{-1} (= mol L^{-1}).

$pH = -\log [H^+] = \log (1/[H^+])$

Lower pH indicates an acidic solution, higher pH an alkaline solution.

At 25 °C, pure water has a pH of 7.

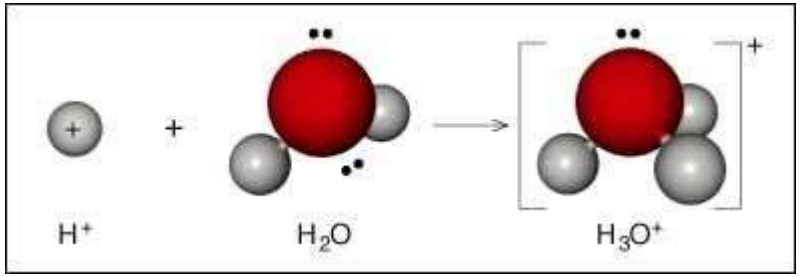
Example: a solution contains 5×10^{-6} mol kg^{-1} hydrogen ions.

$pH = \log (1 / (5 \times 10^{-6}))$

$pH = 5.30$ (mildly acidic)

Misra Ch7.6 and p. 54

In fact, H^+ ions do not exist as isolated ions. They interact with water molecules to form hydronium ions, $H_3O^+_{(aq)}$



Equilibrium constant

- Describes equilibrium in an aqueous reaction

$pH = -\log_{10} [H^+]$

Dissociation reaction for water:

$$H_2O = H^+ + OH^-$$

Equilibrium constant, K for this reaction:

$$K_w = \frac{[H^+]\,[OH^-]}{[H_2O]}$$

Square brackets mean the concentration of the species

Product of species on RHS, divided by (product of) species on LHS

The value of the equilibrium constant at 1 bar and 25 °C (298.15K) is 10^{-14}.

In pure water, the concentration of water is 1 i.e. $[H_2O] = 1$.

$$K_w = [H^+]\,[OH^-] = 10^{-14}$$

At 1 bar and 25 °C (298.15K), $[H^+] = [OH^-] = 10^{-7}$ i.e. pH = 7 (neutral)

Misra p. 54

Acidity: The acidity of a solution is its capacity to donate protons.

A strong acid is dissociated almost completely when dissolved in water.
Examples: hydrochloric acid HCl, nitric acid HNO_3, sulphuric acid H_2SO_4 .

A weak acid dissociates to a small extent, i.e. only a small percentage of its H^+ ions is released into solution.
Examples: acetic acid CH_3COOH, carbonic acid H_2CO_3, phosphoric acid H_3PO_4, silicic acid H_4SiO_4 .

Alkalinity: The alkalinity of a solution is its capacity to accept protons.

Strong bases: hydroxides of alkali and alkaline earth elements
Examples: caustic soda NaOH, slaked lime $Ca(OH)_2$

Weak bases: NH_4OH, $Mg(OH)_2$, $Fe(OH)_3$ (NH_4^+ = ammonium ion)

In an acid-base reaction, the acid is the *proton donor* and the base is the *proton acceptor*.

Dissociation reactions

Acid-base reactions can be written as equations, e.g.

$HCl = H^+ + Cl^-$ (Reaction a)

and

$NaOH = Na^+ + OH^-$ (Reaction b)

We define an equilibrium constant called a *dissociation constant* for each of these reactions:

$$K_a = \frac{[H^+]\,[Cl^-]}{[HCl]} \qquad K_b = \frac{[Na^+]\,[OH^-]}{[NaOH]}$$

Higher values of the dissociation constant indicate a higher degree of dissociation of the acid or base.

Square brackets mean the concentration of the species

Product of species on RHS, divided by (product of) species on LHS

Misra Ch7.3

In the general case, for reactions that have different ratios of cations and anions in the acid / base:

$$H_mA_n = mH^+ + nA^- \quad \text{(Reaction a)}$$

and

$$B_pOH_q = pB^+ + qOH^- \quad \text{(Reaction b)}$$

the dissociation constants are:

$$K_a = \frac{[H^+]^m\,[A^-]^n}{[H_mA_n]} \qquad K_b = \frac{[B^+]^p\,[OH^-]^q}{[B_pOH_q]}$$

For example:

$$H_2SO_4 = 2H^+ + SO_4^{2-} \qquad Mg(OH)_2 = Mg^{2+} + 2OH^-$$

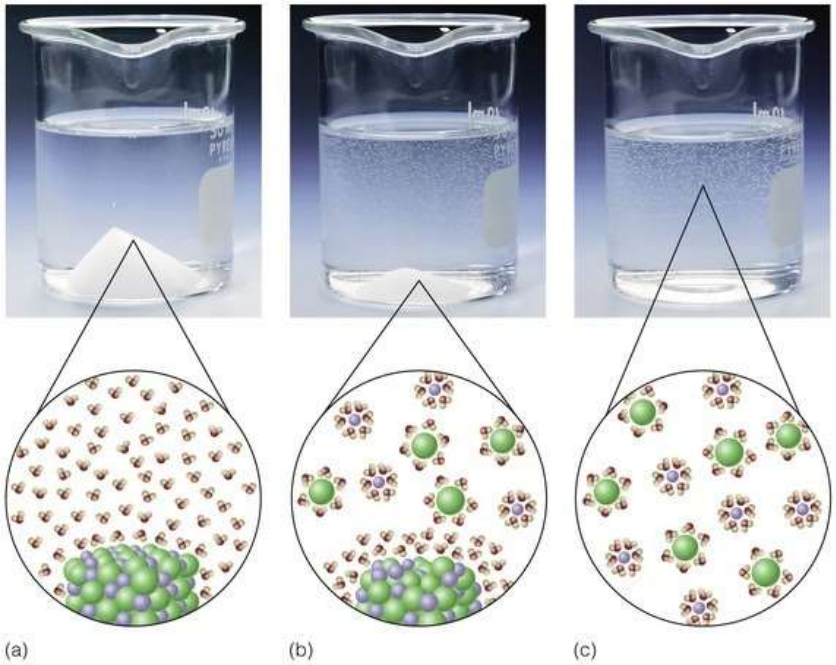
$$K_a = \frac{[H^+]^2\,[SO_4^{2-}]}{[H_2SO_4]} \qquad K_b = \frac{[Mg^{2+}]\,[OH^-]^2}{[Mg(OH)_2]}$$

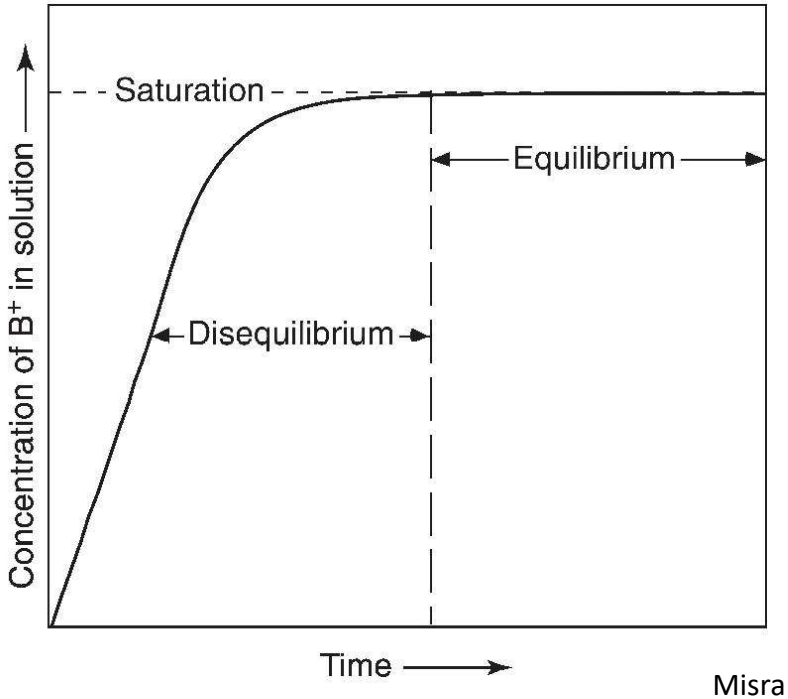
Solubility of Salts

Back to salts: A **SALT** is an ionic compound that contains a cation other than H^+ and an ion other than OH^- or O^{2-} .

The degree to which a salt dissolves in a solution is called its **SOLUBILITY**.

The **solubility** of a substance in water is the concentration of the substance in the water that is in equililbrium with the substance at the temperature and pressure of interest. Such a solution is called a **saturated solution** because it cannot hold any more of the salt.





Misra Fig. 7.5

Misra Ch7.4, p 140-141

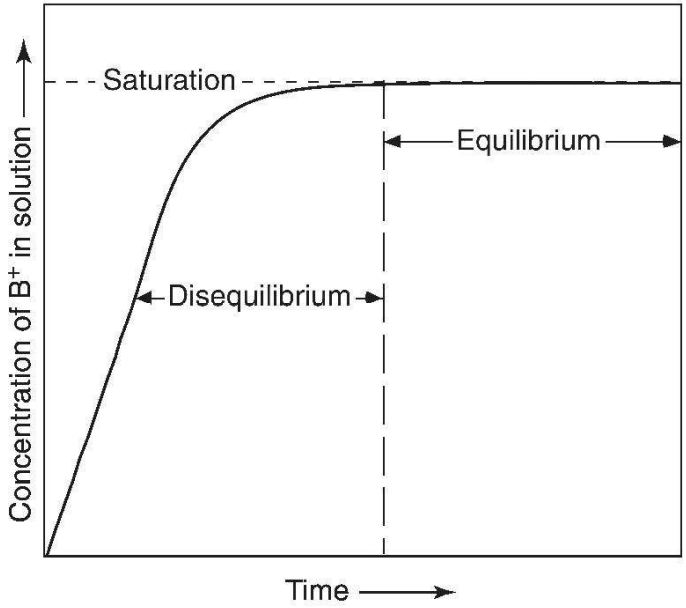
Solubility product

This is a way of expressing the **solubility** of a salt.

For example, for the mineral barite, barium sulphate,

$$BaSO_{4\,(s)} = Ba^{2+} + SO_4^{2-}$$

The **solubility product,** $K_{sp} = [Ba^{2+}]\,[SO_4^{2-}]$



For halite (sodium chloride),

$$NaCl_{(s)} = Na^{+} + Cl^{-}$$

$$K_{sp} = [Na^{+}]\,[Cl^{-}]$$

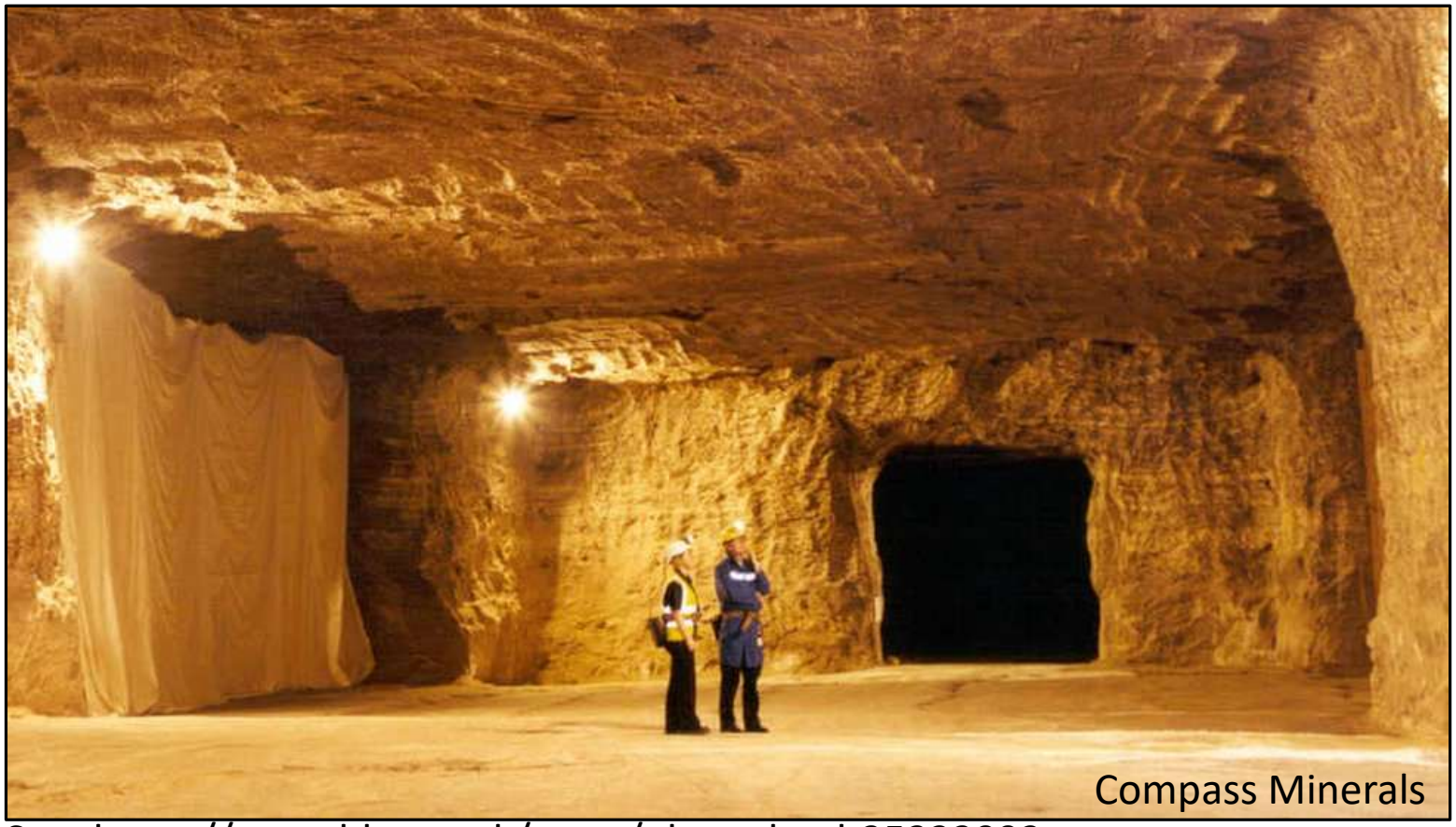
Example: The value of K_{sp} for halite = 36.

This means that for solid halite in equilibrium with water, the concentration of Na (in mol L^{-1}) times the concentration of Cl (in mol L^{-1}) equals 36.

$K_{sp} = [Na^{+}]\,[Cl^{-}] = 36$ so $[Na^{+}] = [Cl^{-}] = \sqrt{36} = 6$ mol L^{-1}

The solution is **saturated** when the concentration of NaCl equals 6 mol L^{-1}.

Rock Salt Deposits



See: https://www.bbc.co.uk/news/uk-england-35322992

Wikipedia: “Cheshire is a county in North West England. Rock salt was laid down in this region some 220 million years ago, during the Triassic period. Seawater moved inland from an open sea, creating a chain of shallow salt marshes across what is today the Cheshire basin. As the marshes evaporated, deep deposits of rock salt were formed.”

The evaporating water reached saturation, resulting in precipitation of halite

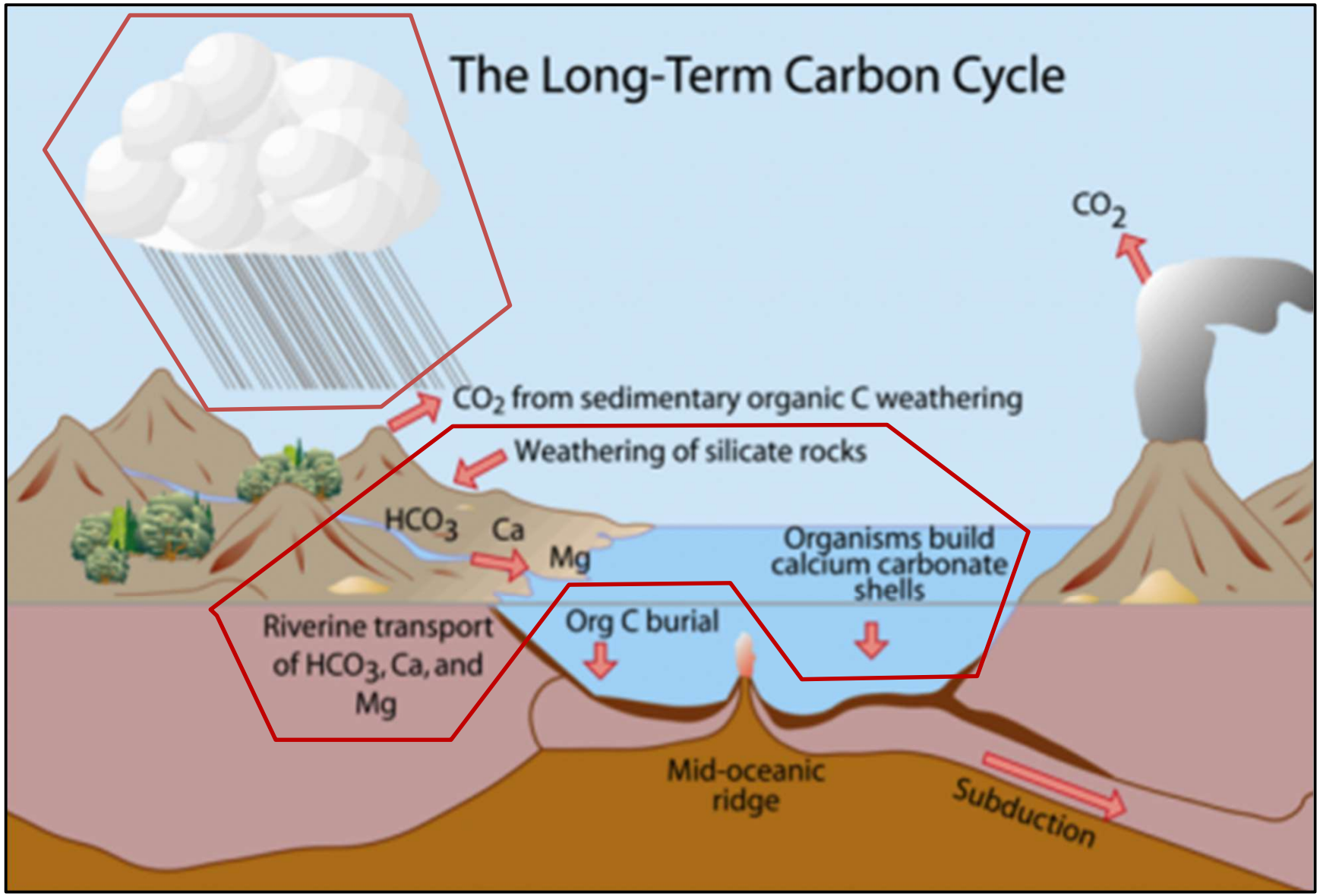
Chemistry of the Earth: Practical 9, Water



Chemical Weathering

Practical 9: Water

- Familiarization with the way chemical reactions are written for aqueous (electrolyte) solutions
- Understand why natural waters can be acidic
- Examine chemical weathering reactions to understand how rocks at the Earth's surface break down through chemical mechanisms


The Long-Term Carbon Cycle
CO_2
CO_2 from sedimentary organic C weathering
Weathering of silicate rocks
HCO_3
Ca
Mg
Organisms build calcium carbonate shells
Org C burial
Riverine transport of HCO_3, Ca, and Mg
Mid-oceanic ridge
Subduction

WATER CHEMISTRY

Acidity

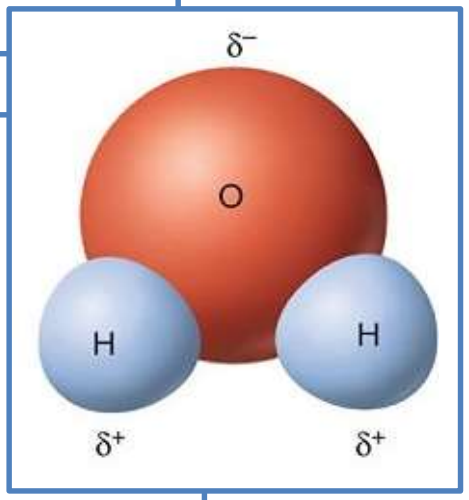
$pH = -\log [H^+] = \log (1/[H^+])$

Solubility product

For halite (sodium chloride),

$$NaCl_{(s)} = Na^+ + Cl^-$$

$$K_{sp} = [Na^+]\,[Cl^-]$$



Equilibrium Constant

For the reaction: $CO_{2\,(g)} + H_2O_{(l)} = H_2CO_{3\,(aq)}$

The equilibrium constant K is:

$$K = \frac{[H_2CO_3]_{(aq)}}{[CO_2]_{(g)}[H_2O]_{(l)}} \qquad K = \frac{A}{B * C}$$

Equations for equilibrium constants can be rearranged just like any algebraic equation:

$$[H_2CO_3]_{(aq)} = K_1 * [CO_2]_{(g)} * [H_2O]_{(l)} \qquad A = K * B * C$$

CARBONIC ACID AND CHEMICAL WEATHERING

Carbonic acid – an important naturally occurring acid that plays a critical role in chemical weathering of rocks.

CO_2 in the atmosphere dissolves in water to form carbonic acid:

$$CO_{2\ (g)} + H_2O_{\ (l)} = H_2CO_{3\ (aq)}$$

Carbonic acid is a weak acid. It dissociates in two steps:

$H_2CO_{3\ (aq)} = H^+ + HCO_3^-$ Bicarbonate: HCO_3^-

$HCO_3^- = H^+ + CO_3^{2-}$ Carbonate: CO_3^{2-}

- Rainwater in equilibrium with atmospheric CO_2 is acidic.
- Higher levels of CO_2 make rainwater more acidic.

Other acids in the atmosphere contribute to "acid rain", Including:

SO_2 gas →→ H_2SO_4 (sulphuric acid)

NO_x gas →→ HNO_3 (nitric acid)

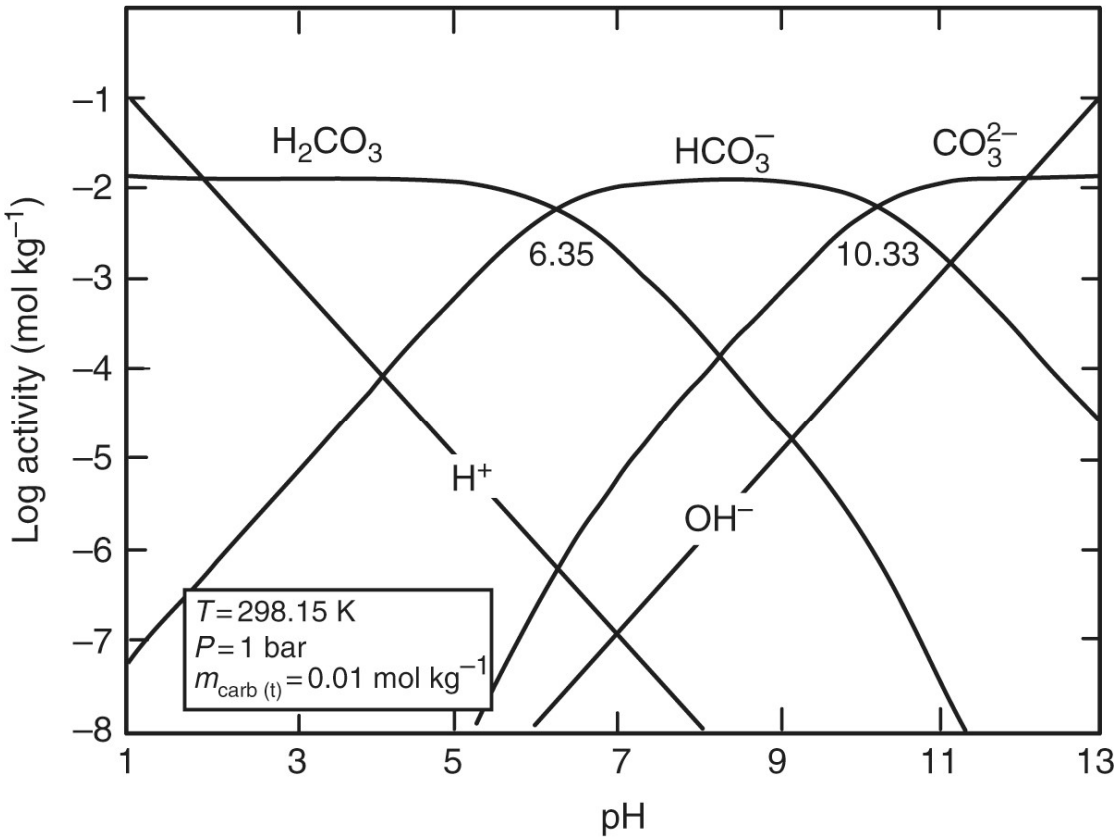


Lovecz | Wikimedia.org

Misra Ch7.5 and p. 53

In the practical:
Activities of dissolved species in the CO_2-H_2O system as a function of pH. Total dissolved carbonate = 0.01 mol kg^{-1}.

Activity ~ Concentration



(Activity takes into account the fact that the chemical activity of an ion is not exactly the same as the concentration, because chemical interactions in the solution can modify the availability of the ion for chemical reactions. Activity is a measure of the "effective concentration" of an ion.)

Misra Fig. 7.6

CHEMICAL WEATHERING

Chemical weathering is the process by which minerals at Earth's surface interact with water (particularly acidic water) and the gases in the atmosphere (CO_2 and O_2) to form mineral assemblages that are thermodynamically more stable under the ambient pressure and temperature (P-T) conditions.

Acids are produced from biogenic activity (plant activity, bacterial metabolism) as well as acidic rainwater.

Chemical weathering can involve :

- **dissolution** reactions e.g. dissolution of halite, carbonates
- **hydration** reactions e.g. formation of gypsum

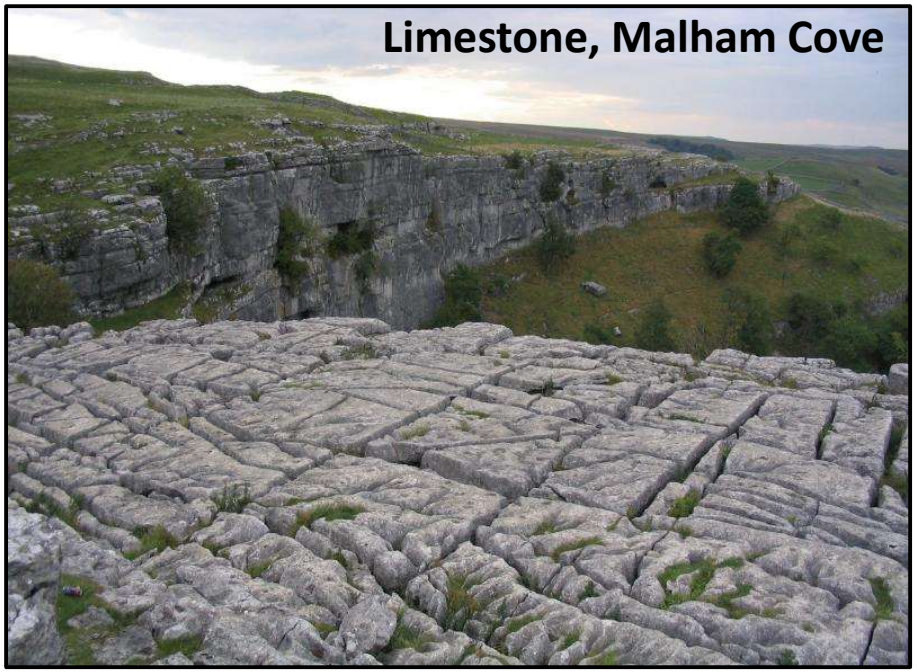
Common *products* of chemical weathering reactions are

- clay minerals e.g. kaolinite, smectite formed in **hydrolysis** reactions
- oxides, hydroxides e.g. hematite, goethite, gibbsite formed in **oxidation** reactions

Misra Ch7.9: Intro and Ch 7.9.1

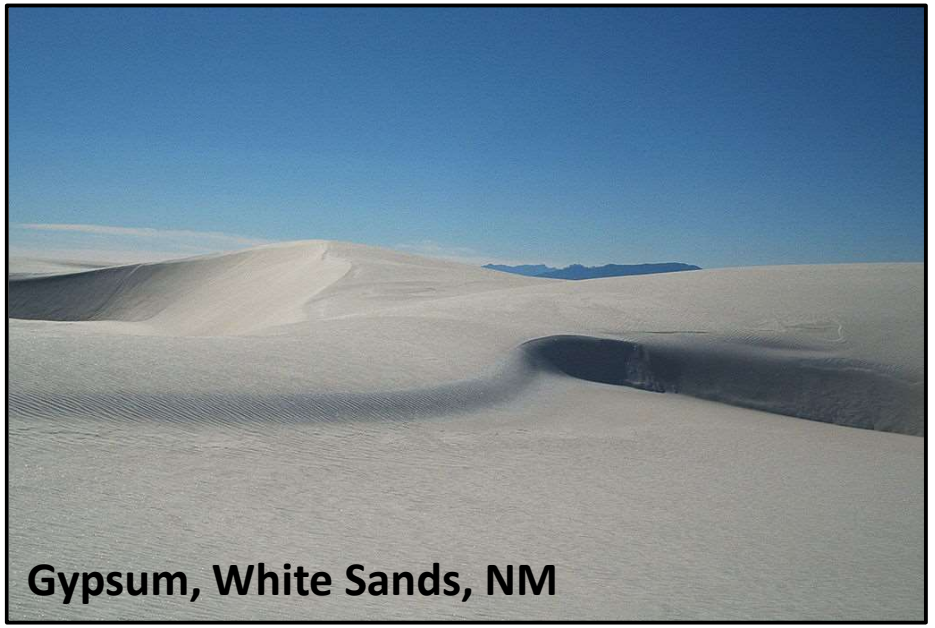
Chemical weathering

Dissolution and carbonation



Wikipedia: Lupin

Hydration



Wikipedia

Oxidation



Hydrolysis



http://en.academic.ru/dic.nsf/enwiki/10221